

Candidate readings of the Xiongnu record

Forty readings of Chinese-transcribed Xiongnu names and titles as Turkic words, with the historical parallels that suggested each one. Each reading priced against the measurements in the paper. Supplied for the record; not offered as results.

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Supplement to *Reading steppe names: what a transcription channel can and cannot recover from the Chinese record of the Xiongnu*

S1 · author_proposals.csv · compiled 2026

S1.1 What this supplement is

Why the file is worth having, given that none of it is a finding. A reading of a Xiongnu name has been offered many times over a century, and offered as a resemblance: this Chinese spelling looks like that Turkic word. What is new here is not the readings but that each one is *priced*. Every row sets out what its reading would require a scribe to have done, and gives the rate at which the corpora show scribes doing it, so that a reader who doubts a reading can see which step to attack and what it would cost to defend. The same treatment is applied to proposals that are not ours, and it is applied to this author without exemption: several readings here were withdrawn while the file was being written because a measurement ruled them out, and §S1.14 records which. A reading that survives that is still not a finding. But it is a different object from a resemblance, and that difference is what the file is for.

Section 9 of the paper sets out what a proposed reading of a Xiongnu name has to survive before it can be called a finding: the sense has to be attested in a source independent of the comparison, the morphology has to be licensed by a suffix with the right distribution, and any objection that can be tested has to be tested. The file covers all forty items of the record, and they are not all of one kind. **One** reading is argued in the paper itself and meets its standard: 若鞮 as *Īnakt* (§9). **One** carries a comparison taken from the literature rather than made by us, which the paper then tests: 徑路 as *Kingrak* (§9.1). **One** is a compound pointing at its constituent rows. **Two** now carry no reading at all. At 羯 the proposal was withdrawn and the reading in print left standing; at 呼都而尸道皋若鞮 it was withdrawn on three measurements and nothing put in its place, and the word it had claimed stays at 烏累若鞮, where the transcription supports it. The remaining **thirty-five** are the author's own proposals and none of them meets the standard, 冒頓 among them: §9.2 of the paper measures the three values its second character can take and does not choose, and the proposal made here is set out in S1.4 with its objections.

They are here since we produced them over the course of the work. A reader who has seen 稽粥 written out as *kei çuk* will hear Turkish *küçük* whether or not this supplement exists. What can be done is to write them down, date them, label them, and put next to each one the reason it occurred to us, so that the reasoning is inspectable rather than implied.

Nothing in the paper depends on this supplement. The measurements in §§5–8 (the channel accuracies, the period-gap decomposition, the initial-consonant probe, the coda-spelling count) were made without reference to these readings and do not change if every one of them is rejected.

A dagger, †, on a syllable in the tables below marks a **polyphonic character**, one for which Schuessler's table gives more than one Later Han reading. The reading printed is the first tabulated one, as in Appendix A of the paper. Table S7 lists every such character with the readings not used, and where an alternative would change a proposed reading the entry for that item says so.

S1.2 Why they are not in the paper

Section 6.3 measures exactly the operation that produced this supplement, taking a reconstructed reading and looking it up in a Turkic dictionary, against a null model built by resampling Later Han syllables into pseudo-names. Across seven lexicons, from modern Turkish to the *Codex Cumanicus*, between none and three names in thirty-six meet significance of $p \leq 0.05$. At a one-in-twenty threshold, thirty-six coin flips give the same.

Doing the matching by hand does not improve on the algorithm. It makes it worse. A person cannot help preferring the reading that means something, and that preference is precisely the selection effect the null exists to quantify. The algorithm at least fails honestly. A paper that measured the unreliability of this operation and then presented its own outputs as discoveries would be incoherent.

So we keep the two things apart. The paper reports what can be measured. This file records what we imagined while measuring it. Because it is supplementary, it does not carry the paper's evidential standard and should not be read as though it did.

S1.3 How the readings were produced

Every row starts from the same two published inputs, neither of them model output. The Chinese form is from the *Shiji*, *Hanshu* and *Hou Hanshu*. The Later Han reading is Schuessler's. The third column simply respells that reading in Turkish orthography, so that a reader who does not read phonetic notation can see what is being claimed. It adds no information.

The fourth column is ours. We arrive at it by hearing the third column as a Turkic word or compound, usually with a vowel changed, sometimes treating a consonant as lost, and then looking for a historical parallel that would make such a name plausible for a steppe ruler. The fifth column gives the sense that reading would carry. The note beneath each row gives the parallel. Each entry below is set out in two parts: *Phonetics*, which walks the Later Han reading to the proposed name character by character and says which of the steps are ordinary correspondences and which are not; and *Reason*, which gives the word, its attestation where it has one, and the naming parallel that suggested it.

The standard we used for inclusion here is deliberately lower than the paper's, and it is worth stating. A reading is offered when an attested word fits the transcription and nothing in the transcription stands against it. It is not offered when the spelling rules it out, when a character would have to carry a sound the corpora never show it carrying. Under-determination is not a reason to withhold a reading, because almost every short name in the record is under-determined; a demonstrable phonological conflict is a reason.

We had three reasoning habits. We list them because they are the main weakness. The first is the epithet argument: later Turkic rulers are commonly named by a common noun or a natural force: Bayezid I *Yıldırım* "the thunderbolt", Alp Arslan "the lion", Mehmed IV *Avcı* "the hunter", so a reading that yields such a noun is treated as more likely. The second is the career argument: where the Chinese record says what a ruler did, a reading that names the deed is preferred. The third is the compound argument: a long transcription is split into two or three Turkic elements. Each of these makes a reading easier to reach, and none of them makes it more likely to be right. They multiply the search space, which is the opposite of evidence.

S1.4 The reading argued in the paper, and this author's proposal for 冒頓

Two items in this supplement are set out at length. The first is argued in the paper rather than merely recorded, because it is supported differently from everything else in the file: the recurring element 若鞮 *śak te*, which closes the formal names of six later chanyu, read *Īnakt*, "the faithful ones" (§9):

- the *Hou Hanshu* glosses 若鞮 as 孝 "filial", the meaning is recorded in the source, not inferred from a resemblance;
- *inak* is attested as *fidelis* "faithful" in the *Codex Cumanicus* (Kuun 1880, p. 182), a medieval source with no connection to the Chinese record;
- the morphology is licensed: *-t* is an Old Turkic plural used with titles (*tegin* → *tegit*, *tarqan* → *tarqat*), and the competing *-tI* reading we ruled out by counting: of 31 Cuman headwords in *-tI*, almost all are past-tense verbs;
- we tested the obvious objection to the spelling and it failed: if the source ended in *-t*, why did the scribe use the open character 鞮? Because 30% of stop-final source syllables in the Sanskrit corpus were written with an open character.

One objection stands and is stated in the paper: if the source word was *inak*, the initial *ɪ-* is not written, and 若鞮 has no room for it.

The second is not a reading the paper argues. 冒頓 *mək tuən^c* is the founder's name, and §9.2 prices the three values its second character can take without choosing among them. **This author's proposal is *Bögü Tuğ***, "sage standard", and it is set out here rather than there, with the objections to it, because that is where proposals in this work belong.

- **The frame.** Both codas are written, which makes it one of the tightest frames in the record, but neither character has a single value. 冒 has two Later Han readings, *mək* and *mou^c*; 頓 has *tuən^c* and *tuət*, and a third value from the commentarial note 頓音毒, which Pulleyblank reports as Early Middle Chinese *dawk*. That third value is a substitution recorded by a commentator, not a row of the table, and the paper labels it as such.
- **The first element.** 冒 and 默 share the reading *mək*, and 默啜 is read *Bögü-Çor*, the personal name of Qapaqhan Qaghan; 牟 *mu* writes the same word in 牟羽 for the Uyghur *Bögü* Qaghan. So both rows of 冒 have a precedent for writing *bögü*. Table 9 of the paper prices them: a rounded source vowel is written with a rounded Chinese vowel in 59% of cases and an open one in 4%, so the second row is where *bögü* is ordinary rather than merely permitted.
- **The second element.** *tuğ* is the horse-tail or yak-tail standard carried on a pole, one of the oldest continuously used objects of steppe rulership. It is attested in Old Turkic: *tuğ taşıkar* "when the standard went out" in the Moyun Çor inscription, cited by Doerfer at ETY I 170, and Kāšgarī glosses it as the drum and banner borne before the king, with *toquz tuğluğ xan* "a khan of nine standards". It ends in a velar, so it needs 頓's second row (2 of 21 opportunities) or the substitution (10 of 21), and it is excluded on the first row (0 of 21).
- **The count that follows.** Two of the three values of 頓 admit a velar-final second element, one admits the liquid-final *Bayatur* the field has read for a century, and one admits a nasal-final one. That is the whole of the arithmetic, and it is why this author prefers a velar-final reading.
- **An observation about the substitution.** In the Later Han table 毒 and 蠱 are exact homophones, *douk*, and 蠱 is the Chinese word for the yak-tail standard itself. So the commentator who wrote 頓音毒 put the value of 頓 at precisely the Later Han reading of the word for the standard. This may be coincidence, and one homophone pair proves nothing, but it is recorded here rather than left out.
- **The name shape.** *bögü* is a regnal epithet across the record: *Teŋri Bögü Teŋriken* in the Old Turkic inscriptions, *Bögü* Qaghan of the Uyghurs (牟羽), *Bögü-Çor* of the Türks (默啜). Kāšgarī glosses *bügü* as "bilgin, akıllı, hakim"; Clauson says the word "seems to connote both wisdom and mysterious spiritual power", and it was borrowed early into Mongolian as *böge*, the ordinary word for a shaman. Two nouns in apposition is the commonest Turkic name pattern.

The objections, all of which stand.

- **The loan direction.** Doerfer's *TMEN* gives Turkic *tuğ* as a loan from Chinese (entry 969). If that is right, the word cannot be an element of a name of the early second century BC, and the reading fails. Doerfer's own addendum in volume IV weakens it: it quotes Menges (1968: 169, as cited there) that the Chinese word "might well be an ancient loanword in Chinese", and H. Franke, in a letter of 23 March 1966, that the Chinese word is first attested in the *Zhouli* in a purely Chinese context and is "kaum jünger als 2. Jh. v. Chr.", which is the founder's own century. The direction is therefore disputed by the same authority usually cited for it, and this author does not settle it.
- **The row it needs.** The reading is ordinary on the substitution, marginal on 頓's second row, and excluded on the first, which is the row Appendix A of the paper prints. It also reads better on the second row of 冒 than on the first, which Appendix A likewise prints.
- **The compound is not attested.** Only its two halves are, as with every compound reading in this supplement.

- **The frame does not fix the vowel** of the second syllable, so *tuǰ* is what the velar coda and the Turkic lexicon together suggest, not what the transcription requires.

The rival reading is not free of these either, and it is fair to say so. Clauson's entry on *bayatur* records that the word "occurs only once in the early period and then still as a P.N.", that the sense "picked warrior" most probably developed in Mongolian rather than Turkic, and that Turkic had other words such as *alpayut* in that meaning; Doerfer's entry 817 adds that the form *bayatur* is not findable in the Old Turkic dictionaries and texts available to him, the attested Old Turkic form being *bātur*, and that the reconstruction **bayatur* is secured by Old Hungarian *bahatur* rather than by a Turkic attestation. A form without a medial velar leaves 冒's written coda with nothing to write.

Both appear in the tables below in their proper places, so that this supplement can be read as one list with the standard visible in it. The second appears there as an author's proposal, which is what it is.

S1.5 Why the decoder's output is not cited below

The channel of §4 has a decoding step that will return a ranked list of readings for any string in the record. None of those rankings is quoted in this supplement, and the reason is worth stating, because the numbers look more informative than they are.

A character can be *reading-anchored*, meaning its emission distribution was learned for that character specifically, or it can fall back to a table keyed on the Later Han **initial consonant class alone**, discarding the vowel and the coda. Of the 39 items in the record, only five are reading-anchored throughout; 34 use backoff on at least one character and nine use it on every character. The effect is that whole groups of characters share one distribution:

LATER HAN INITIAL CLASS	CHARACTERS IN THE XIONGNU RECORD SHARING ONE EMISSION DISTRIBUTION	WHAT THE BACKOFF TABLE RETURNS FOR EVERY ONE OF THEM
velar	17: 稽 渠 胡 湖 壺 戶 侯 姑 孤 骨 谷 韓 諧 皋 權 胸 徑	ka 0.34, ga 0.15, ku 0.08
affricate	6: 且 粥 珠 臣 詹 郅	ca 0.18, ja 0.10, cā 0.08
dental stop	9: 鞮 題 當 屠 撐 株 稚 脫 頓	ta 0.23, da 0.11, ti 0.10
liquid	6: 廬 孿 祿 累 蠡 閭	ra 0.37, la 0.17, rī 0.08

TABLE S1 Because 稽 *kʰei*^a and 渠 *giə* draw on the same row, and 粥 *tʃuk* and 且 *tʃə*^a on another, an affricate-plus-velar string and a velar-plus-affricate string return mirror images of one table entry rather than anything specific to those characters. Two further limits apply: the scores are normalised over a retained beam, so they are relative rankings within it and not posteriors; and the syllable inventory is Sanskrit-derived, so *q* and the front rounded vowels *ö* and *ü* cannot be produced at all, so a reading containing *ü* is therefore not rejected by the decoder, it is unavailable to it. §6.2 of the paper makes the general point; this table is its mechanism.

In the entries below, where a spelling question arises, we cite the paper's initial-correspondence counts (§8.1): direct counts of which Chinese initial was used for which source sound across the three corpora. Those are observations rather than model output, and they are what the word "permitted" means wherever it appears in this supplement.

S1.6 Chanyu, in order of reign

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
頭曼 Touman	do man ^{c†}	do man	Duman	dust cloud
PHONETICS 頭 <i>do</i> → <i>du</i> : the initial is direct; o and u are not separable at the ~50% vowel recovery §7 reports. 曼 <i>man</i> → <i>man</i> : exact, and the coda -n is written, because -n is one of the six codas the Chinese syllable can carry. One vowel change in two syllables. The rival reading <i>tūmen</i> needs <i>ü</i> , which the Later Han script cannot write at all. REASON <i>duman</i> / <i>tuman</i> / <i>touman</i> "smoke, fog, mist" is one word across Kāşgarî, the Codex Cumanicus and modern Turkish. The standard identification is <i>tūmen</i> "ten thousand", which is absent from the Cuman vocabulary entirely.				
冒頓 Modu / Modun	mək [†] tuən ^{c†}	mık tuın	Bögü Tuğ	sage standard

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS The frame is b/m – k/g – t/d – nasal or stop, and both codas are written, which makes it one of the tightest frames in the record and also one of the least determinate, because neither character has a single value. 冒 has two Later Han readings, <i>mæk</i> and <i>mou</i>; 頓 has <i>tuən</i> and <i>tuət</i>, and a third value from the commentarial note 頓音毒, reported by Pulleyblank as Early Middle Chinese <i>dawk</i>, which is a substitution recorded by a commentator rather than a row of the table. 冒 <i>mæk</i> → <i>bögü</i>: the m- stands for a source b-, 3 of 22 in the Later Han corpus, 13.6%; the written -k stands for the source g; and the final vowel is not written, one character covering two source syllables, a device found in 47 of the 1,110 verified pairs, as 竭 covers <i>-gata</i> in <i>tathâgata</i>. There is a character-level precedent as well as a rate: 冒 and 默 share the reading <i>mæk</i> and 默啜 is read <i>Bögü-Çor</i>, while 牟 <i>mu</i> writes the same word in 牟羽 for the Uyghur Bögü Qaghan. 頓 → <i>tuğ</i>: a velar-final second element, which the corpus writes with the -t coda of 頓's second row in 2 of 21 opportunities and with the -k of the substitution in 11 of 23, and never with the -n of the first row, 0 of 21. REASON Bögü Tuğ, “sage standard”. <i>tuğ</i> is the horse-tail or yak-tail standard carried on a pole, attested in Old Turkic (<i>tuğ taşıkar</i> “when the standard went out”, Moyun Çor, cited by Doerfer at ETY I 170) and glossed by Kâşgarî as the drum and banner borne before the king, with <i>toquz tuğluğ xan</i> “a khan of nine standards”. <i>bögü</i> is a regnal epithet across the record: <i>Teñri Bögü Teñriken</i>, Bögü Qaghan of the Uyghurs, Bögü-Çor of the Türks; Kâşgarî glosses <i>bügü</i> as “bilgin, akıllı, hakim” and Clauson says it “seems to connote both wisdom and mysterious spiritual power”. Both halves are attested independently of any Chinese source and two nouns in apposition is the commonest Turkic name pattern. Why a velar-final second element. Of the three values 頓 can take, two admit a velar-final element, one admits the liquid-final <i>Bayatur</i> the field has read for a century, and one admits a nasal-final one. §9.2 of the paper sets out that comparison and does not choose; the preference here is this author's. What is not settled. The loan direction of <i>tuğ</i> is disputed: Doerfer's <i>TMEN</i> 969 gives it as a loan from Chinese, which would put it out of reach of a name of the early second century BC, while his own addendum in volume IV quotes Menges (1968: 169, as cited there) that the Chinese word “might well be an ancient loanword in Chinese” and Franke (letter of 23 March 1966) that the Chinese word is first attested in the <i>Zhouli</i> and is “kaum jünger als 2. Jh. v. Chr.”. A ninth-century check adds nothing either way, and is recorded so that it is not made twice: Tekin's glossary to the <i>İrk Bitig</i> has the verb <i>tug-</i> “to rise (of sun)” but not the noun, the text's word for a flag is <i>uçrug</i>, and <i>bögü</i> is not in it. That text is sixty-five omens and a few hundred words, so absence from it is not evidence of anything. The reading is also ordinary on the substitution, marginal on 頓's second row and excluded on the first, which is the row printed here; the compound is not itself attested, only its halves; and the frame does not fix the vowel of the second syllable. The rival reading carries its own difficulties, recorded in S1.4. Full discussion in S1.4.				
稽粥	k ^h ei ^ʙ † tśuk	kei çuk	Kiçük	the younger
Jizhou / Jiyu PHONETICS 稽 <i>k^hei</i> → <i>ki</i>: an aspirated velar is the ordinary vehicle for a source k, and the diphthong <i>ei</i> for <i>i</i> is routine. 粥 <i>tśuk</i> → <i>çük</i>: tś- is the default character class for a source affricate (37% in the Sanskrit corpus), and the coda -k is actually written, the strongest part of the reading, since a stop coda goes unwritten 30% of the time. Only ü is unavailable and surfaces as u. REASON <i>kiçig</i> / <i>kiçik</i> / <i>kiçük</i> “small, younger” is attested three times independently: Kâşgarî has <i>kiçik</i> and <i>kiçük</i> with the derivative <i>kiçiglik</i>, the <i>Kutadgu Bilig</i> has <i>kiçig oğlan erken</i> “when he was a small boy” (1823), and the Codex has <i>Kečak</i> glossed <i>paucus</i>. 稽粥 is the personal name of Laoshang chanyu, Modu's son and successor, so “the younger” fits the succession the Chinese sources record.				
軍臣	kun džin [†]	kun cin	Köngen	the one who is always upright
Junchen PHONETICS 軍 <i>kun</i> → <i>kön</i>: three steps, all direct. A source velar written with a Chinese <i>k-</i> is 204 of 236, 86.4%; a source rounded vowel written rounded is 207 of 324, 63.9%; the -n is written, 94 of 111, 84.7%. 臣 → <i>-gen</i> takes the second of the character's two Later Han readings, the velar <i>gin</i>, rather than the palatal <i>džin</i> that Appendix A prints: a source <i>g</i> written with a Chinese <i>g-</i> is 64 of 110, 58.2%, and a source front vowel written with a front vowel is 349 of 406, 86.0%. The palatal reading is not available to this name. A source voiceless affricate written with the voiced <i>dž-</i> is 0 of 53 in the Sanskrit corpus, so <i>çin</i>, <i>çen</i> and <i>çan</i> are excluded outright; the Codex forms that appear to offer an affricate are an artefact of medieval Latin spelling, in which <i>j</i> writes Turkic <i>y-</i> and <i>c</i> before a back vowel writes <i>k-</i>. Step 47 prices every combination the two characters permit, and <i>köngen</i> is the cheapest at 23.4%, about 1 in 4, against 1 in 9 for the best branch of 冒頓 and the 1 in 37 that retires <i>Bayatur</i> in §8.1 of the paper. Two syllables, two characters, nothing inserted and nothing discarded. REASON Köngen, “the one who is always upright”. The stem is the verb <i>kön-</i>, which Kâşgarî glosses <i>düzelmek</i>, <i>doğrulamak</i>; <i>yola gelmek</i>, to straighten, to become upright, to come to the right way, and which carries a family around it: <i>könitmek</i>, <i>köndgermek</i> and <i>köndgürmek</i>, all <i>doğrultmak</i>. The suffix is <i>-gAn</i>, the deverbal participle with a habitual force. The suffix is counted here, not assumed. Across the 9,379 headwords of the Kâşgarî index, the Codex Cumanicus and Tekin's glossary to the <i>İrk Bitig</i>, <i>-gAn</i> has 211 headwords and 109 of them are glossed with <i>daima</i>, <i>her zaman</i> or <i>çok</i>: <i>arıtgan</i> “her zaman temizleyen”, <i>eğilgen</i> “daima eğilen”, <i>etilgen</i> “her zaman düzelen”, which is the near sibling of the form proposed here. The competing suffixes <i>-gUn</i> and <i>-gIn</i> have 72 headwords between them and not one habitual gloss: they are concrete nouns, birds and plants and body parts, a knot, a bride. In the eleventh-century record <i>-gAn</i> is the productive participle and <i>-gUn</i> is not. The modern <i>durgun</i>, <i>çoşkun</i> and <i>ötgün</i> belong to a later productivity of <i>-gUn</i> and are not evidence for this period. Vowel harmony decides a tie the transcription cannot. <i>Köngin</i> costs exactly what <i>köngen</i> costs, 1 in 4, because 臣 carries a front vowel either way. But <i>-gAn</i> has two shapes while <i>-gIn</i> and <i>-gUn</i> have four, so a front rounded stem such as <i>kön-</i> must take <i>-gün</i> and cannot take <i>-gin</i>. The Turkic evidence removes a candidate that the Chinese evidence cannot. What is not settled. <i>Köngen</i> is not itself an attested headword. It is a formation by a productive rule from an attested stem, which is a stronger position than a form requiring a segment that must afterwards be lost, but it is not an attestation. The lexicons are eleventh century and the name is second century BCE, so that <i>-gAn</i> was productive for Kâşgarî does not establish it 1250 years earlier. Whether <i>-gAn</i> formations were used as personal names or titles in Old Turkic is not tested here. The <i>g</i> to <i>g</i> rate is measured over all positions, and whether a <i>g</i> after a nasal behaves the same has not been measured. And the first element is undetermined: <i>kön</i>, <i>kün</i>, <i>kon</i> and <i>kun</i> all cost the same 47.0%, so choosing <i>kön-</i> is a choice made on meaning, which is the first of the three reasoning habits this supplement lists against itself.				
詹師盧	tśam ši lia	çam ši lia	Çamşılı	dazzling, radiant
Zhanshilu PHONETICS 詹 <i>tśam</i> → <i>çam</i>: tś- is the default affricate spelling and the -m coda is written. 師 <i>ši</i> → <i>şı</i>: the retroflex sibilant is the standard vehicle for ś. 盧 <i>lia</i> → <i>lı</i>: note that Chinese l- also carries source r-: in the paper's correspondence counts an r- was written l- in 12 of 14 attestations, so this syllable is genuinely ambiguous between -lı and -rı. REASON A stem of shining or dazzling. The solar reading this once echoed at 軍臣 has been withdrawn, so it stands alone.				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
响犁湖 Xuli hu / Gouli hu PHONETICS 响 <i>huo</i> → <i>ka</i>, 犁 <i>lei</i> → <i>li</i>, 湖 <i>ga</i> → <i>k</i>: three onsets, each taken from the majority correspondence, with nothing inserted and no coda discarded. The velar class covers h-, k- and g- alike, so the first and third syllables cost nothing. REASON Kāşgarî glosses <i>kalık</i> “hava, gök, sema” (air, sky, heaven). The word belongs to the same semantic field as 撐犁, the one Xiongnu term the Chinese sources gloss (天 “heaven”), so a ruler bearing it would not be isolated in the record. A second exact fit, <i>kılık</i> “disposition, manner”, is available if the sky reading is thought too convenient.	huo[†] lei[†] ga	huo lei ga	Kalık	sky, heaven
伊稚斜 Yizhixie PHONETICS 伊 <i>ʔi</i> → <i>i</i>: a glottal onset is the standard way to write a vowel-initial syllable. 稚 <i>ɟi</i> → <i>di</i>: retroflex ɟ for d is routine. 斜 <i>ja</i> → <i>ya</i>: this character class emits <i>ya</i> more strongly than any other single-syllable correspondence in the record. All three steps are default. REASON <i>idi</i> “lord, master”, attested in Old Turkic. Compare Murad I's epithet <i>Hüdavendigâr</i>, “the sovereign”, a title of the same shape and force.	ʔi ɟi^c ja	’i di ya	İdiya	master
烏維 Wuwei PHONETICS 烏 <i>ʔa</i> → <i>a</i>: vowel-initial, direct. 維 <i>wi</i> → <i>bɪ</i> is the weak step: the Chinese character carries a w-, and while Turkic b- and w- alternate, the transcription reads more naturally as <i>avɪ</i> than <i>abɪ</i>. This is one of only five items in the record decoded entirely from character-specific evidence, so the mismatch cannot be attributed to a gap in the data. REASON <i>av</i> “hunt”, <i>avcı</i> “hunter”. <i>Avcı</i> names an occupation; <i>abɪ</i> is proposed as naming a trait. <i>Avcı</i> is also the epithet of Mehmed IV.	ʔa wi	’a vi	Abı	one who hunts or stalks
且鞮侯 Qiedihou PHONETICS 且 <i>tʂa</i> → <i>ça</i>: ts- for a source affricate is a minority spelling but a real one, 15% in the Sanskrit corpus and 9% in Ligeti. 鞮 <i>te</i> → <i>tr</i>: direct. 侯 <i>go</i> → <i>gka</i> asks a single Chinese syllable to carry a consonant cluster it structurally cannot, so the reading adds material the transcription does not have. REASON <i>çat-</i> “to strike together, to clash”. Compare Selçuk Bey, whose name has been read as “perpetually ready for the fight”.	tʂa^B te[†] go	tʂa te go	Çatıgka	one ready for the clash
狐鹿姑 Hulugu PHONETICS 狐 <i>ɣua</i> → <i>kur</i>: a velar fricative for a Turkic back <i>q</i>, which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character; <i>kurluga</i> has back vowels throughout. The rounded vowel is written rounded, 207 of 324, 63.9%. The <i>-r</i> is not written, and that is the ordinary treatment rather than a difficulty: Table 7 gives a source liquid coda spelled with an open character 21 times in 37, 56.8%, the majority option, and 狐 is open. 鹿 <i>lok</i> → <i>lu</i>: a source <i>l</i> written with a Chinese <i>l</i>- is 97 of 100, 97.0%, the vowel rounded again. Its written -k is not discarded. The character after it is 姑 <i>ka</i>, a velar, and a medial written coda writes the onset of the following syllable in 165 of 446 cases, 37.0%, which is the largest single thing such a coda does: 僧 for <i>sam-ghā</i>-, 三 for <i>sam</i>-, 怛 for <i>ta-thā</i>-. 姑 <i>ka</i> → <i>-ga</i>, with the vowel open, 1127 of 1634, 69.0%. The whole name costs 2.00%, about 1 in 50, against the 1 in 37 that retires <i>Bayatur</i> in §8.1. The weak step is the last one, not the first two. A source <i>g</i> written with a Chinese <i>k</i>- is 23 of 110, 20.9%. Were the word <i>kurluka</i> the same character would be writing a source <i>k</i> at 86.4% and the name would cost 1 in 12, so one consonant moves the price fivefold. Step 51 sets this out. REASON <i>Kurluga</i>, from the sense of binding and enclosing. The root is attested and so is the sense: Kāşgarî has <i>kur</i> “kuşak, kemer”, a belt or girdle, and the verb <i>kurlamak</i>, “kuşak yapmak ve bağlamak”, to make a belt and to bind, together with <i>kurluk</i> and <i>kurlanmak</i>. 狐鹿姑 captured the Han general Li Guangli and held the confederation together through a hard decade, so an epithet of binding fits what the record says he did. Compare the Göktürk ruler Kapgan Kagan, <i>kapgan</i> “the encloser”. Two objections in an earlier draft of this entry were wrong and are withdrawn. It described the <i>-r</i> as a segment Later Han cannot write and the <i>-k</i> as a discarded coda, and said the two pulled in opposite directions. Neither holds: leaving a liquid unwritten before an open character is the majority spelling at 54%, and a written coda reaching forward to the next onset is the commonest use of a written coda at 37%. Both were judged by ear before they were counted. What is not settled, and it is the shape of the word rather than any of its sounds. <i>Kurluga</i> is not an attested headword, and neither is <i>kurluka</i>; the frame returns no word at all. More than that, the shape is not one Turkic builds: the suffix <i>-IXk</i> / <i>-IXg</i> is word-final, 380 headwords end in it, while only one headword in 9,343 ends in <i>-luka</i> and none in <i>-luga</i>. The well-formed Turkic word in this frame is <i>kurluk</i> or <i>kurlug</i> “girded”, which 狐鹿 write between them with 鹿's own <i>-k</i> as the final consonant, and which leaves 姑 unaccounted. That is the same structural question as at 虛閭權渠: whether a longer Xiongnu name is one word or a name followed by a title element. It is not answered here, and this reading assumes the first.	ɣua lok ka	ğua lok ka	Kurluga	the binder, the encloser
壹衍鞮 Huyandi PHONETICS 壺 <i>ga</i> → <i>ka</i>: given a Later Han voiced g-, the source sound is more often k than g, so the devoicing is normal rather than a liberty. 衍 <i>jan</i> → <i>yan</i>: direct, coda written. 鞮 <i>te</i> → <i>tr</i>: direct. All three steps are default or better, phonetically one of the cleaner readings here. REASON <i>kay-</i> “to slide, to slip away”. Parallel in sense to Bayezid I's <i>Yıldırım</i>: a natural force taken as a regnal epithet.	ga jan^B te[†]	ga yan te	Kayantı	avalanche
虛閭權渠 Xulüquanqu	k^hia lia gyan ɣia	kia lia gyan ɣia	Kalkan	shield

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 虚 <i>kʰia</i> → <i>ka</i> , 閭 <i>lia</i> → <i>l</i> , 權 <i>gyan</i> → <i>kan</i> : the first three characters match exactly, including the written -n coda, at zero minority steps. Only 渠 <i>gia</i> is left unaccounted for, and it may carry a separate title element. REASON <i>kalkan</i> “shield” is in Kāšgarī and in the Codex Cumanicus as <i>Kalkani</i> , “scutum ejus”. A weapon or armour name is an ordinary steppe formation, and it needs no compound to be read.				
握衍胸鞬 Woyanqudi	ʔɔk jan ^{B†} guo te [†]	’ok yan guo te	Okyan Kutuǵtı	blessed sovereign of the arrow flank
PHONETICS 握 <i>ʔɔk</i> → <i>ok</i> is exact, with the -k coda written, and 衍 <i>jan</i> → <i>yan</i> and 鞬 <i>te</i> → <i>ti</i> are direct. The weak point is 胸 <i>guo</i> → <i>kutuǵ</i> , which expands one Chinese syllable into two Turkic ones; nothing in the corpora licenses that expansion. REASON <i>ok</i> “arrow” doubles as a division of the confederation in later Turkic usage (the <i>On Ok</i> , “ten arrows”, of the Western Turks) which is what makes a territorial reading available. <i>kut</i> is “heavenly favour, fortune”.				
呼韓邪 Huhanye	ha [†] gan ja [†]	ha gan ya	Kaganya	kagan of the flank
PHONETICS 呼 <i>ha</i> → <i>ka</i> : a source back <i>q</i> written with an <i>h</i> - character, which §8.5 of the paper measures at 43 of 66, 65.2%, against 20 of 66 for an aspirated character. It is the majority spelling, not a substitution needing indulgence. The 27% figure earlier drafts quoted here was the Mongolian rate; the Turkic one is the relevant measurement and it is higher. 韓 <i>gan</i> → <i>gan</i> : exact, coda written. 邪 <i>ja</i> → <i>ya</i> : a source <i>y</i> - takes this character class 57 times in 98, 58.2%, the largest single destination for a source <i>y</i> - though not a majority of the corpus. The written sequence <i>ha gan ya</i> becomes <i>ka-gan-ya</i> on one change. REASON 呼韓邪 commanded the left flank before becoming ruler, so the Chinese record itself supplies the office that the reading names.				
鄯支 Zhizhi	tʃit tʃe [†]	çit çe	Çitçi	border-maker, one who carves the boundary
PHONETICS 鄯 <i>tʃit</i> → <i>çit</i> : tʃ- is the default affricate vehicle and the -t coda is written. 支 <i>tʃe</i> → <i>çi</i> : default. Both syllables use the majority correspondence and neither requires an inserted or deleted segment. REASON <i>çit</i> / <i>çiz</i> -, of marking or carving a boundary. 鄯支 broke away westward and set up a rival seat, so a name of division fits the career, though it is equally possible that the career suggested the name.				

TABLE S2 The fifteen rulers whose names are not built on the recurring element 若鞬. Reign order follows the *Shiji* and *Hanshu* as reproduced in Appendix A of the paper.

s1.7 The six 若鞬 compounds

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
若鞬 Ruodi	ńak te [†]	nyak te	İnakt	the faithful ones
PHONETICS 若 <i>ńak</i> → <i>ınak</i> : a palatal nasal onset with the -k coda written. 鞬 <i>te</i> → <i>-t</i> : the final stop is not written, and Table 7 of the paper licenses that at 30%. One objection stands and the paper states it: if the source was <i>ınakt</i> , the opening <i>ɪ</i> - is not written and 若鞬 has no room for it. REASON One of the two readings argued in the paper itself, at §9, and it is listed here so this supplement can be read as a single list with the standard visible in it. <i>ınak</i> is glossed <i>fidelis</i> “faithful” in the Codex Cumanicus; the <i>Hou Hanshu</i> glosses 若鞬 as 孝 “filial”; the plural in <i>-t</i> is an Old Turkic formation used with titles (<i>tegin</i> → <i>tegit</i>); and Clauson records Chagatay <i>ınak</i> as “a royal representative or senior minister”, so a court title on this root is what the word ordinarily does.				
復株累若鞬 Fuzhulei Ruodi	buk [†] ʧio lui [†] ńak te [†]	buk tio lui nyak te	Büktel İnakt	Büktel, of the ınakt
PHONETICS 復株累 <i>buk ʧio lui</i> → <i>büktel</i> : the b-, the -k coda, the dental and the liquid all fall in place across the three characters, at zero minority steps, with only the front rounded vowel unwritable. One attested word covers the whole prefix rather than an invented two-element compound. 若鞬 → <i>ınakt</i> is treated in §S1.4. REASON Kāšgarī gives <i>büktel</i> as “orta boylu; yassı arkalı, oturamaklı”: of middling height and flat-backed. A name from a bodily description is a well-attested Turkic type; compare the Hungarian royal name <i>Şişman</i> .				
搜諧若鞬 Souxie Ruodi	so [†] gei ńak te [†]	so gei nyak te	Sarığ İnakt	the yellow, the fair, of the ınakt

			READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY		
PHONETICS 搜 <i>so</i> → <i>sar</i> : a sibilant onset, which the correspondence counts make obligatory: a source k, q or g was written with a Chinese sibilant in 0 of 2,124 attestations, while Chinese sibilant characters transcribe s, š, č or ś in 98% of 837 cases. The final -r is not written and could not be: Later Han carries no -r in any coda position. 諧 <i>gei</i> → <i>ig</i> : a velar onset, direct. Nothing is inserted, and no character is read as a sound it cannot carry. 若鞮 → <i>ínakt</i> is treated in §§1.4.				
REASON <i>sarig</i> “sari, sarı renk” (yellow) is in Kâşgarî, and survives as modern Turkish <i>Sarı</i> , still current both as a name and as an epithet. Color epithets are a documented Turkic naming type: the same dictionary has <i>çakır</i> , which Kâşgarî glosses <i>azraq</i> “blue” and applies to eyes in his own proverb, <i>it çakırı atqa tegir</i> , <i>at çakırı itqa tegmes</i> , “a blue-eyed dog is worth a horse, a blue-eyed horse is not worth a dog”, because such a horse sees badly; the derived verb <i>çakrat-</i> is glossed “to roll the eyes and look blue-eyed”. Modern Turkish <i>çakır</i> is greyish-blue of the eyes, and earlier drafts of this entry glossed it that way; the eleventh-century gloss is the plain colour. <i>Çal</i> “dappled, grey” is proposed for 烏珠留若鞮 on the same footing. Ninety-one attested headwords fit this two-syllable frame, so it does not determine the reading; <i>sarig</i> is offered as the most plausible of them, with nothing in the transcription standing against it.				
車牙若鞮 Cheya Ruodi	tʃʰa [†] ɲa ɳak te [†]	ça nga nyak te	Tunga <i>ínakt</i>	the tiger, or the hero, of the <i>ínakt</i>
PHONETICS 車 <i>tʃʰa</i> → <i>to</i> : the first of the character’s two tabulated readings, no polyphony argument needed. A source <i>t</i> written with the aspirated <i>tʃʰ</i> - is 67 of 198, 33.8%; the character is open, so any source coda would be unwritten; the vowel is free, a source <i>o</i> written with an open vowel being 103 of 324, 31.8%. 牙 <i>ɲa</i> → <i>-ɲa</i> : a source syllable beginning with a velar nasal takes a Chinese <i>ɲ</i> - character 5 times in 7, and the one intervocalic case is 丁彦多 for <i>tiñanta</i> , where 彦 carries the <i>-ñan</i> -; a source <i>a</i> written with an open vowel is 1127 of 1634, 69.0%. The whole name costs 5.29%, about 1 in 19, against the 1 in 37 that retires <i>Bayatur</i> in §8.1 of the paper. Step 48 prices every form the two characters permit. The reading this replaces is excluded twice over. <i>Ok</i> required 牙 to write a vowel-initial syllable, which is 0 of 140 in the Sanskrit corpus, 137 of the 140 taking a glottal; and <i>çöge</i> required a source affricate written with the aspirated <i>tʃʰ</i> -, which is 1 of 59, where the vehicle for a source affricate is the plain <i>tʃ</i> - at 37 of 53. It also added a syllable the character does not carry. A cheaper form exists and is rejected on sense, not on phonology. On the second reading of 車 the frame admits <i>kaña</i> , from the Codex <i>canga</i> , at about 1 in 3; it is glossed <i>tabulatum</i> , planking, which is not a ruler’s name, and it needs the second reading where <i>toña</i> needs only the first. The transcription under-determines here as it does everywhere, and the choice is made on meaning, which §6.3 of the paper measures as the weakest step in the whole procedure. On the spelling. The <i>-ng</i> - of Turkish <i>Tunga</i> renders a single velar nasal, not a nasal followed by a stop. That is why these two characters write the word and 當戶 cannot: a written <i>-ɲ</i> coda followed by a velar onset is the corpus device for a source <i>ɲg</i> or <i>ɲk</i> cluster, 13 cases of 13, and this word has no velar stop in it. REASON <i>Tunga</i> , an epithet, in the title slot the other 若鞮 names use. Clauson’s headword is <i>toña</i> -, with back vowels, and he records that “whether -o- or -u- in the first syllable is uncertain”, so <i>Tunga</i> and <i>Tonga</i> are equally available; this file prints the first, and the transcription cannot separate them either, since 車 carries an open vowel and a source <i>o</i> and a source <i>u</i> fall alike at 31.8%. Two senses, resting on different evidence, and both are recorded here. Kâşgarî glosses the word as a tiger-like beast (<i>Divân</i> III 368), and glosses the title <i>Toña Alp Er</i> , given to Afrasiyab the king of the Turks, as “the heroic man as strong as a tiger”. Outside personal names the eighth- and ninth-century Uyghur texts use it for “hero, outstanding warrior”: the Manichaean <i>éligler xanlar toñalar</i> , “kings, khans, great warriors” (M III 41), the Buddhist <i>Arcunı toña</i> , “the hero Arjuna” (U II 24), and <i>toñalar begi</i> . Clauson records that the animal sense “is not confirmed by any other authority”, and adds that by Kâşgarî’s day the word’s meaning had become unknown. That second point rests on an emendation, and the two standard editions of the <i>Divân</i> disagree. The manuscript reads <i>salla</i> , and Dankoff and Kelly translate it as it stands: “this word has remained with the Turks and its meaning persists among them.” Clauson prints “its meaning became unknown” and marks the change in his own parenthesis, <i>read dalla for salla in MS</i> . Whether the animal sense was live or already fossil in 1072 therefore turns on one letter, and it does not affect this reading either way: on both readings the word was in use as a title, which is the only thing this name needs. The naming pattern is attested, and it is the one this name needs. Kâşgarî says the word is <i>yulaqqab bihi</i> , used as an honorific epithet, gives <i>toña xan</i> and <i>toña tegin</i> , and glosses the title itself, <i>toña alp ar</i> , as “a man, a warrior, as strong as a tiger”. All of this is taken from Dankoff and Kelly’s edition rather than at second hand. The shape is the epithet first, the title after it, head last, which is the shape of <i>Alp Arslan</i> and <i>Bilge Kağan</i> . <i>Tunga Ínakt</i> is that shape, with 若鞮 as the title element the paper argues in §9, and it matches the form the other five 若鞮 rows already take, such as <i>Çal Ínakt</i> . Both word orders are Kâşgarî’s, and they belong to different registers. The verse he quotes runs <i>alp ar toña öldi mü / esiz ajun kaldı mu</i> , “has Afrasiyab died, has the wretched world been left behind”, which is the order familiar in Turkish as <i>Alp Er Tonga</i> . The title is the other way round, <i>toña alp ar</i> , and so are <i>toña xan</i> and <i>toña tegin</i> . Clauson cites the title form and the Turkish tradition remembers the verse form; neither is mistaken. This bears on the reading, because 車牙若鞮 stands in a title slot, and there the <i>toña</i> -first order is the attested one. What is not settled. The vowel of the first syllable, which neither Clauson nor the transcription fixes. Whether the epithet is the animal or the quality. Both senses are recorded above, neither is suppressed, and the transcription cannot choose between them. And this word stood at 當戶 in earlier drafts; it is moved here because 當戶 writes a source nasal-plus-velar cluster and this word has a single nasal.				
烏珠留若鞮 Wuzhuliu Ruodi	ʔa tʃo liu ɳak te [†]	’a ço liu nyak te	Çal <i>ínakt</i>	the grey or dappled one, of the <i>ínakt</i>
PHONETICS 烏 <i>ʔa</i> is vowel-initial, so the reading begins at 珠 <i>tʃo</i> → <i>ç</i> , with 留 <i>liu</i> → <i>l</i> : an affricate written with the default <i>tʃ</i> - character and a liquid, both majority correspondences. Nothing is inserted, and no character is read as a consonant it does not carry. REASON <i>çal</i> is “alaca, kır” in Kâşgarî: dappled, grey. Color epithets are a persistent Turkic naming pattern, as <i>çakır</i> shows in the same dictionary: Kâşgarî glosses it <i>azraq</i> “blue” and applies it to eyes in a proverb about a blue-eyed dog and a blue-eyed horse. He also records <i>Çaqır Toña Xan</i> as a ruler’s name, so the colour word is attested both as a gloss and in use as a name element.				
烏累若鞮 Wulei Ruodi	ʔa lui [†] ɳak te [†]	’a lui nyak te	Uluğ <i>ínakt</i>	the Great, of the <i>ínakt</i>

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 烏 <i>ʔa</i> → <i>u</i>:- a source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140, 97.9%, which is the strongest single correspondence used anywhere in this file. 累 <i>lui</i> → <i>-lu</i>: a source <i>l</i> is written with a Chinese <i>l</i>- 97 times in 100, 97.0%, and the vowel is rounded on both sides, 207 of 324, 63.9%. The final <i>-ǰ</i> is not written, which the coda table of §8 puts at 9 of 23, 39.1%, the ordinary treatment of a source velar coda. Two characters, two syllables, nothing inserted and nothing discarded. The element costs 23.7%, about 1 in 4. That figure is not comparable with a whole-name figure such as the 1 in 4 for <i>Köngen</i> or the 1 in 37 that retires <i>Bayatur</i> in §8.1: it prices a two-syllable element, and a short string is cheap by construction, since every further character multiplies in another rate below 1. 若鞮 is not priced again here, being the one title element §9 argues once for all six rows that carry it. REASON <i>Uluǰ</i> “great”, from the Codex Cumanicus <i>Oulu</i>, glossed <i>magnus</i>. The word is among the most common elements in Turkic titlature, which is both the strength of this reading and its weakness: a very common word is an easy target to hit, and §6.3 of the paper measures choice by meaning as the weakest step in the whole procedure. The same word was read at 呼都而尸道皋若鞮 in earlier drafts and is withdrawn there. Step 54 measured what separates the two rows. This one has a glottal character for the vowel-initial syllable and a liquid character for the <i>-l</i>-; the long name has neither, and no character in it carries a liquid at all. Where one word appears twice in a record, the transcription and not the sense has to say which occurrence is real, and here it does. What is not settled. Whether the word is a title, an epithet or part of a personal name. The record gives no way to tell, and the other five 若鞮 rows are equally silent on it.				
呼都而尸道皋若鞮	ha [†] ta ná [†] ší dou ^{ʙ†} kou nák te [†]	ha ta nyɪ ši dou kou nyak te		
Huduershi Daogao Ruodi PHONETICS The reading earlier drafts gave this name, <i>Öz-Uluǰ İnakt</i>, is withdrawn, and none is proposed in its place. It fails three measurements, none of them close. First, twice over, on the zero this file uses elsewhere to retire a reading. A source syllable beginning with a vowel takes a Chinese glottal onset 137 times in 140; it takes a velar 0 times in 140 and an <i>h</i>- 0 times in 140. Both <i>öz</i> and <i>uluǰ</i> begin with a vowel, and the characters asked to write them are 呼 <i>ha</i> and 皋 <i>kou</i>. The same zero retired <i>Çöge-Ok</i> at 車牙若鞮. Second, on the liquid. A source <i>l</i>- is written with a Chinese <i>l</i>- 97 times in 100, 97.0%, and a source liquid onset of either kind, <i>l</i> or <i>r</i>, takes a Chinese <i>l</i>- 262 times in 288, 91.0%. Not one of the eight characters carries a liquid onset, so the <i>-l</i>- of <i>uluǰ</i> would have to go unwritten in the one position the corpus almost always writes it. Third, on the four characters left over. Reading 呼 as <i>öz</i>, 皋 as <i>uluǰ</i> and 若鞮 as the title leaves 都 <i>ta</i>, 而 <i>ná</i>, 尸 <i>ší</i> and 道 <i>dou</i> carrying nothing. Across 1,110 pairs in the Sanskrit corpus no transcription uses more than one character beyond the syllables of its source: 772 are one to one, 152 use one extra, 93 use fewer, and the count of pairs with four or more spare characters is 0. Step 50 established that most of those single extras are a Chinese semantic suffix appended to a finished transcription, 鬼鳥國城, which these four are not. The reading cannot be priced, and that is the result. A zero has no rate to multiply. REASON No reading is proposed here. This is the longest string in the record and the one where a reading is easiest to invent, because the longer the string the more ways there are to cut it into Turkic elements, and the cut then does the work that no individual correspondence is doing. That is the failure mode §6.3 of the paper measures, and it is what produced the withdrawn reading. <i>Uluǰ</i> stays at 烏累若鞮, where the transcription supports it. What the characters do carry. Step 54 searched the lexicons at the classes the first six characters actually hold, 呼 a velar or <i>h</i>, 都 a dental stop, 而 a palatal nasal, 尸 a sibilant, 道 a dental stop, 皋 a velar. Every two-character frame returns hits and none of them reads as a ruler's name; they are listed in the step output as a starting point for someone else, not as candidates. What is not settled. What the name says. The Chinese sources gloss none of it, the string is long enough that the record cannot constrain a reading of it, and nothing here is evidence that it is or is not Turkic.				

TABLE S3 The recurring element itself, then the six later chanyu whose formal names end in it. Only the shared element is argued for in the paper; the material preceding it in each name is a proposal like any other in this supplement.

s1.8 The ruling clan name, in its two written forms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
攀鞮 Luandi	lyan te [†]	lyan te	El-indi	state founder
PHONETICS 攀 <i>lyan</i> → <i>el</i> moves the liquid from the onset of the Chinese syllable to the coda of the Turkic one; no corpus shows that kind of metathesis. 鞮 <i>te</i> → <i>indi</i> expands one syllable into two. The reading is driven by the sense rather than by the transcription. REASON <i>el</i> “realm, state” with the definite past in <i>-di</i>: literally “he founded the realm”. This is the clan name as the <i>Shiji</i> and <i>Hanshu</i> write it.				
虛連題 Xulianti	k ^h ia lian ^{ʙ†} de [†]	kia lian de	Kıyalıǰ El-indi	state founder of the rocky region
PHONETICS 虛 <i>k^hia</i> → <i>kıya</i> is reachable. 連 <i>lian^ʙ</i> → <i>liǰ</i> adds an unwritten coda, and 題 <i>de</i> → <i>el-indi</i> carries a whole two-syllable word on one character. Three characters are being asked to yield five syllables. REASON The same clan name as 攀鞮, in the form the <i>Hou Hanshu</i> gives, with the <i>kaya</i> element recurring from 虛閭權渠.				

TABLE S4 攀鞮 in the *Shiji* and *Hanshu*, 虛連題 in the *Hou Hanshu*. Both are taken here as the same name, which is the usual view.

S1.9 Titles and lexical items

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
單于 chanyu	džan [†] wa	can va	Çarvuş	army commander; the earlier form of çavuş
PHONETICS Two characters, 單 with a written coda and 于 open. 單 → <i>çar-</i> takes the second of the character's two Later Han readings, the dental <i>tan</i>, for the onset: a source voiceless affricate written with the voiced <i>dž-</i> of the first reading is 0 of 53 in the Sanskrit corpus, while with a dental onset it is 8 of 48, 16.7%. For the coda it takes the Old Chinese value, Baxter and Sagart's <i>*dar</i>, whose liquid becomes the <i>-n</i> of the later table; step 43 measures how often the two reconstructions differ in this way across the record. 于 <i>wa</i> → <i>-vuş</i> is the best value this character offers: a source <i>v</i> written with a Chinese <i>w-</i> character is 9 of 65, 13.8%, against 1 of 55 for a source <i>b</i> and 0 of 140 for a vowel-initial syllable. The final <i>-ş</i> is unwritten, which a sibilant coda is in 27 of 36 cases, 75%, since Later Han has no sibilant coda. The liquid is what makes the form possible. Across the 8,930 headwords of the Kâşgarî index and the Codex, <i>-rv-</i> and <i>-rb-</i> occur 18 times in native words (<i>burba-</i>, <i>karva-</i>, <i>kurbaka</i>, <i>turbi</i>), while <i>-nv-</i> occurs 0 times and <i>-nb-</i> 5 times, all in loanwords: a liquid before a labial is Turkic, a nasal before one is not. Two shapes the frame excludes outright, whatever word is tried. A velar first element is not available: a voiced affricate writing a velar stop appears once in 220 in the Sanskrit corpus and that one case is a translation mislabelled as a transcription, and characters that write <i>kan</i> exactly, 干 <i>kan</i> and 汗 <i>gan</i>^c, were available and not used. A vowel-initial second element is not available either: a glide character writing a vowel-initial source syllable is 0 of 140. REASON Çarvuş, the earlier form of çavuş, “army commander”. A native Turkic title of the first rank. Clauson does not mark it foreign, unlike <i>beg</i>, <i>xan</i> and <i>xağan</i>, and Doerfer 1055 has it travelling outward, to Persian and thence to Urdu and Lezgian. It is attested in the eighth-century inscriptions in the highest possible company: <i>bilgesi çavuşı erti</i>, “he was his Counsellor and Army Commander” (Ix. 17), and at Tonyukuk 7 the speaker is Tonyukuk himself. Clauson glosses it “originally the principal military officer of a xağan, the military counterpart of <i>bilge</i>”, and adds that “as time went on the post gradually lost status”, ending as the modern Turkish sergeant; Kâşgarî defines the office as the one who orders the ranks in battle and restrains the troops from outrages (I 368), and his form of the word is <i>çaβuş</i>, with a bilabial fricative. Why it is proposed here. Founders of dynasties commonly bear military titles: Osman Gazi, Alp Arslan, Alp Tegin. And a supreme title that began as a military one is an ordinary trajectory, as with Latin <i>imperator</i> “commander” and <i>dux</i>. The Türk inscriptions are a thousand years later than the Xiongnu, so the decline Clauson describes was already under way when they were cut, and at the earlier stage the office Clauson calls the principal military officer of a xağan could have been the ruler's own title rather than his officer's. What is not settled. Çarvuş itself is not recorded in the later Turkic sources. Clauson's <i>Etymological Dictionary</i>, Erdal's <i>Grammar of Old Turkic</i>, Kâşgarî's <i>Dîwân</i>, the <i>Kutadgu Bilig</i>, Doerfer's <i>TMEN</i> and Tekin's edition of the <i>Irğ Bitig</i> all carry çavuş or çaβuş and none carries a form with the liquid. On this reading the <i>-r-</i> was lost before the eighth century, which is a thousand years after the name was written down.				
撐犁孤塗單于 chengli gutu chanyu	ḍaŋ lei [†] kua da džan [†] wa	dang lei kua da can va	(compound of the three rows below)	
PHONETICS Six characters forming the full ceremonial title, and the only item in the record the Chinese sources gloss in full. It decomposes into three units that each have their own row here: 撐犁 <i>ḍaŋ lei</i>, 孤塗 <i>kua da</i> and 單于 <i>džan wa</i>. No separate reading is proposed for the compound; whatever the three parts are read as, the whole follows. REASON The <i>Hanshu</i> glosses the phrase as “Son of Heaven, the Chanyu”, with 撐犁 = 天 “heaven” and 孤塗 = 子 “son”. That makes it the single most valuable item in the record, because both the form and the meaning are recorded, and it is the evidence on which most of the language-attribution argument has rested for a century.				
撐犁 chengli	ḍaŋ lei [†]	dang lei	Tangli	bearing divine light; sacred
PHONETICS 撐 <i>ḍaŋ</i> → <i>tang</i>: retroflex ḍ for t is routine and the <i>-ŋ</i> coda is written. 犁 <i>lei</i> → <i>li</i>: direct. Both steps are default. The same two syllables equally support <i>tengri</i>, which the Chinese gloss 天 and the lexicon search of §6.3 both favour: Kâşgarî returns it 4th, the Codex 2nd. REASON <i>tang</i> of dawn or light, read as “bearing divine light; sacred”. Offered as an alternative to the standard identification with <i>tengri</i>, not as an improvement on it.				
孤塗 gutū	kua da	kua da	Kuti	the blessing of heaven, with the third-person possessive

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 孤 <i>kua</i> → <i>ku</i>: the default velar onset, direct, with the coda open. 塗 <i>da</i> → <i>ti</i>: a dental onset carrying the <i>-t</i> of <i>kut</i>, with the vowel free. Two characters, two syllables, nothing inserted and nothing discarded. The frame is k/g/h – t/d and both codas are unwritten, so a source <i>-r</i>, <i>-l</i>, <i>-s</i> or <i>-z</i> could hide in either. 66 attested headwords fit it. REASON <i>kuti</i>, “the blessing of heaven”, with <i>-i</i> as the third-person possessive. <i>kut</i> is the divine favour that legitimises a ruler: Clauson records it in the hendiadys <i>kut kiv</i> “divine favour”, and it is the word the <i>Kutadgu Bilig</i> is named from. The possessive is not padding but what Turkic requires of the second noun in a genitive-possessive phrase, <i>tengri kuti</i>, “the <i>kut</i> of heaven”, which is the shape of the full title 撐犁孤塗單于. The phrase is attested, not assembled. Clauson’s entry for <i>kut</i> cites the <i>Irk Bitig</i> for <i>tenri kutinta</i> “by the favour of heaven”, which is <i>tenri kuti</i> with a locative on it, and the Orkhon inscriptions for the dative <i>kutıŋa</i>, in the passage where the ruler’s mother is said to have enjoyed the favour of heaven. So both the possessive form and the two-word phrase are on record in Old Turkic, and the Chinese title maps onto them element for element. That is now checkable in the primary edition rather than only in Clauson’s report of it: Tekin’s glossary to the <i>Irk Bitig</i> carries <i>kut</i> as a headword, glossed “divine favor, blessing, good fortune”, and its citations include <i>kutinta</i> at omens 15 and 56, the first of them in the phrase <i>tenri kutinta</i>. The manuscript is ninth-century. In Old Turkic that suffix is <i>-i</i> / <i>-i</i>, so <i>kuti</i> rather than <i>kutu</i>; the <i>-u</i> of modern Turkish <i>kutu</i> comes from labial harmony, a later development. The two-character spelling is itself evidence for the second syllable. Had the word been a bare <i>kut</i>, a single character was available that writes it exactly, 骨 <i>kuət</i>, with the <i>-t</i> on the page, and the record uses that character in 骨都侯. A scribe who had the one-syllable option and wrote two characters was writing two syllables. The vowel of 塗 is <i>a</i> and matches neither <i>i</i> nor <i>u</i> closely, but §7 of the paper puts vowel recovery at 50.0% with the interval straddling one half, so the vowel is the least informative part of the spelling and cannot decide between them. The Chinese gloss is 子 “son”, and this reading does not translate it. That is recorded against the proposal rather than explained away. The word the gloss wants is <i>oġul</i>, and <i>oġul</i> is barred by the writing rather than merely disfavoured: it is vowel-initial while 孤 carries a velar, and in the Sanskrit corpus a source vowel-initial syllable took a zero or glottal onset in 96% of 183 cases and a velar in none of them. Its medial <i>-ğ-</i> has nowhere to sit either, since 塗 is dental. So <i>oġul</i> is not one of the 66 headwords that fit the frame and never was a candidate; the same wall excluded <i>okluk</i> at 胡祿. Of the 66 that do fit, none means “son”: they include <i>kut</i> “heavenly favour, fortune”, <i>kat</i> “layer”, <i>kata</i> “time, occasion”, <i>ketü</i>, <i>Kerti verus</i> and <i>kötü</i>. The Yeniseian rival, as its author now leaves it. Pulleyblank compared the word with Arin <i>bikjäl</i> “son”. In 2000 Vovin held that Arin form to be isolated in the family, the other languages reflecting Proto-Yeniseian <i>*puʔ-b</i> “son” (Ket <i>hiʔp</i>, Yug <i>fiʔp</i>, Kott <i>fup</i>), and found no Altaic etymology either (Vovin 2000, <i>CAJ</i> 44/1, p. 91). He withdrew it three years later: “I would like to take back my suggestion that Arin form is isolated in Yeniseian” (Vovin 2003, Part 2). He then rebuilds the comparison rather than dropping it, reading Pumpokol <i>p-halla</i> “my son” as a cluster with the first-person prefix rather than as an <i>f-</i>, which makes it cognate with the Arin form, and ruling out Pulleyblank’s Ket <i>qal/qaloq</i> instead. The Yeniseian side of this item is therefore stronger than Part 1 alone would suggest, and it is recorded here as he left it. So this is the one item in the record where two independent kinds of evidence point at different words: the writing supports <i>kuti</i> and the gloss asks for a word the writing excludes. The gloss can be accounted for without supposing the informant was confused about Xiongnu. If the title meant “the heaven-favoured chanyu”, the Chinese expression that fits it is 天子, “Son of Heaven”, which is their own imperial title. 天子 has two characters where the Xiongnu title has two words, so the element glosses follow mechanically from that equation: 撐犁 aligns with 天 and 孤塗 with 子. On that account the first gloss is a true lexical gloss, since 撐犁 is <i>tenri</i> “heaven”, and the second is a back-formation from the phrase-level equivalence rather than testimony about what 孤塗 means. This is an explanation of the gloss and not a test of it, which is why the reading is offered here rather than argued in the paper.				
屠耆 tuqi	da gi [†]	da gı	Doğru	straight, upright, honest
PHONETICS 屠 <i>da</i> → <i>do</i>: a dental onset, direct. 耆 <i>gi</i> → <i>ğru</i>: a velar onset, with the <i>-r</i> unwritten, which is forced rather than permitted, since Later Han carries no <i>-r</i> in any coda position. 144 attested headwords fit the frame, so the transcription does little to narrow the choice; the vowels are free as always. REASON This is the author’s proposal, and the comparison is ours. We have not found this reading proposed in print. The search that would have found it covered every book held for this project, including Golden’s <i>Introduction to the History of the Turkic Peoples</i> in a machine-readable form made for the purpose, and returned nothing. It is recorded here as our own comparison and not as a reading taken from the literature. The Codex Cumanicus has <i>Togru</i> glossed <i>rectus</i>. The Chinese sources gloss 屠耆 as 賢 “worthy, wise”, and it appears in 左右屠耆王, the Wise Kings of the Left and Right. The senses meet more closely than “straight” suggests. Clauson derives <i>toġuru</i>: / <i>toġru</i>: as a gerund of <i>toġur-</i>, physically “straight” and then metaphorically “straight, honest, upright, true”, and that moral sense is old rather than modern: a fourteenth-century glossary renders it with Arabic <i>al-tiqa</i> “trustworthy, honest”, and <i>toġru ayt-</i> with <i>šadaqa</i> “to tell the truth”. The word survives across the whole Turkic family and as modern Turkish <i>doğru</i>, “straight, correct”. Toġrıl and Toġrul are not offered in support. The two names resemble <i>toġru</i> in sound, and they are well attested: Kāšġarī records that men are called <i>Toġrıl</i>, an <i>Alp Toġrıl Tégin</i> appears in a Uyghur list of proper names, and the first Seljuk sultan bore the name. But <i>Toġrıl</i> is a bird of prey, and Clauson derives it from <i>toġra-</i> rather than from <i>toġur-</i>, noting that it never survived as a common noun. What makes the two look like one word is a separate homophone, the passive <i>toġrıl-</i> “to be upright”, which is from <i>toġur-</i> and survives as modern Turkish <i>doġrul-</i>. So the names are recorded here to head that resemblance off, not to rest anything on it: the case for <i>Doğru</i> stands on the Chinese gloss 賢 and on the attested moral sense of <i>toġru</i>. The paper does not argue this reading; §9.1 there tests only proposals taken from the literature.				
谷蠡 luli / guli	kok [†] le ^{B†}	kok le	Körklü	glorious, imposing

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 谷 <i>kok</i> → <i>körk</i>: the default velar onset, with the -k coda written and the -r unwritten before it. Later Han has no -r coda, so a source -rk cannot be written at all without spending a further character on an <i>l</i>- syllable; leaving it unwritten is the only option two characters allow, and it is the majority practice: step 31 counts 63 verified pairs whose source form has a liquid before a consonant, and the liquid is unwritten in 54 of them, 86% (達磨 for <i>dharmā</i>, 薩達摩 for <i>saddharma</i>, 播鉢吒 for <i>parpaṭa</i>). In 6 of the 63, 10%, the scribe spent a further character carrying an <i>l</i>- onset instead, so the step this reading asks for has a measured minority against it rather than none. 𪛗 <i>le^B</i> → <i>lū</i>: the <i>l</i>- onset direct, with an open coda. Two characters, two syllables, one unwritten consonant. Both are polyphonic characters: the table gives 谷 as <i>kok</i>, <i>jok</i> and <i>lok</i>, and 𪛗 as <i>le^B</i> and <i>loi^B</i>; this reading takes the first of each. REASON <i>Körklü</i>, “glorious, imposing”, taking the word at the broader end of a range that runs from beauty to splendour. The word is a headword in both lexicons used here, and its sense broadens across the centuries, which the reading has to reckon with. Kāšġarī’s own Arabic is at the narrow end: he glosses <i>körk</i> as <i>ḥusn</i>, <i>jamāl</i>, beauty, and <i>körklüg</i> as <i>jamīl</i>, beautiful, with <i>ḥasan</i> elsewhere. Earlier drafts of this entry gave the gloss as <i>iyi, güzel ve gösterişli</i>, “good, beautiful and imposing”, which is the Türk Dil Kurumu index’s Turkish rendering rather than Kāšġarī’s Arabic, and the “imposing” in it is not his. Clauson’s attestations sit at the same narrow end: <i>körklüg menizliġ</i> “good-looking” in Uyghur Buddhist texts, <i>kuba körklügüġ kôr</i> “see the fair-haired beauty” in the <i>Kutadġu Bilig</i>, and a Kipchak <i>inen körklü</i> “extremely beautiful”. The broader sense appears later, and it is well attested. The <i>Codex Cumanicus</i> glosses <i>körk</i> as <i>decus</i>, glory or dignity rather than prettiness. The <i>Book of Dede Korkut</i>, Oghuz and preserved in fifteenth and sixteenth century manuscripts, carries <i>görklü</i> as a fixed devotional epithet: <i>adı görklü Muhammed</i>, “Muhammad whose name is glorious”, recurs as a formula before battle, and a formulaic catalogue runs <i>ayna günü okunanda hutbe görklü, kulak urup dinleyende ümmet görklü</i>, “glorious is the sermon read on Friday, glorious the congregation that hears it”. In the Deli Dümrl story God is addressed as <i>görklü Tanrı</i>. None of these referents is merely pretty, and the highest of them is God. Modern Turkish continues the same end of the range with <i>görmek</i> “splendour, magnificence” and <i>görmekli</i> “imposing”, both from this stem. Two things keep that from settling the row. The broad sense is Oghuz and late, four centuries after Kāšġarī and far later still than the record this supplement reads, and in <i>Dede Korkut</i> it attaches almost entirely to Islamic religious referents, which is a devotional register rather than a general one. The narrow sense is meanwhile still the current one elsewhere in the same century: the fourteenth century Kipchak vocabularies Clauson cites give <i>körk</i> as <i>al-ḥusn wa-l-jamāl</i> (Houtsma 19, 12) and <i>al-ḥusn</i> (Ibn Muḥannā 80), beauty and nothing further. In the oldest layer <i>körk</i> renders Sanskrit <i>rūpa</i>, “form” (Suv. 164, 20), and the related <i>körtle</i> is a component of women’s names in the Uyghur documents. WE therefore record the broad sense as attested and the narrow one as the older, and price this reading on its form rather than on its gloss. Clauson records the form across the whole span, with Kipchak and Coman <i>körklü</i> / <i>körkli</i> and Osmanlı <i>görklü</i>. The Kipchak form matters for the spelling: <i>körklü</i> carries no final -g, so the open coda of 𪛗 is simply correct and nothing further needs licensing. The word is used of exalted figures and not only of pretty objects: an Old Turkic Buddhist text has <i>tenridem kız azu tenri oġlını teg körkle oġlan</i>, “a child as beautiful as a divine girl or a son of God” (TT V 12, 127, via Clauson). Two caveats belong with that citation: Clauson calls the text a fairly late one, and he queries whether the -le form <i>körkle</i> / <i>körtle</i> is native rather than borrowed. The form this reading uses, <i>körklüg</i>, is not in doubt, being a straightforward derivation in -lüg from the attested <i>körk</i>. An authority assigns 谷 a different reading in this very title. Pulleyblank’s <i>Lexicon</i>, at K1202a, glosses 谷 read <i>lu</i> as “luli king, Xiongnu title”, cross-referring <i>gu</i> and <i>yu</i> as the other readings. That is the <i>lok</i> branch, on which no Turkic word can fill the first syllable at all, so if Pulleyblank is right this reading is not available and 谷𪛗 is not a Turkic word. The proposal is recorded with that against it, unresolved. The other readings of 谷 do not compete. On <i>lok</i>, which the office’s conventional name, the Luli Kings, would suggest, the first syllable begins with a Chinese <i>l</i>-, and a Chinese <i>l</i>- writes a source <i>l</i>- or <i>r</i>-. Turkic admits neither initially: Clauson calls initial <i>r</i>-, like initial <i>l</i>-, a sound entirely foreign to the language, and loanwords that established themselves took a prosthetic vowel. That branch yields no candidates at all, so if 谷 is <i>lok</i> here, 谷𪛗 is not a Turkic word. On <i>jok</i>, which is Schuessler’s palatal glide, the frame is y...k plus <i>l</i>/<i>r</i> and admits <i>yüklüg</i> “laden” and <i>yokla</i>- “to rise, ascend”, neither of which suits an office. What cannot be tested here. The Chinese sources gloss 撐犁 and 孤塗 and say nothing about 谷𪛗, so this row carries no semantic constraint at all, only the rank of the office: the 谷𪛗王 of the Left and Right stood among the four highest, paired with the Wise Kings. The choice among the candidates therefore rests on the epithet argument that §S1.3 names as this supplement’s main weakness, and the reading is offered as permitted rather than confirmed.				
當戶 danghu	tan ⁺ ga ^B	tang ga	Tamga	the ruler’s seal, and the officer who keeps it
PHONETICS 當 <i>tan</i> → <i>tam</i>:- a source <i>t</i> written with a Chinese t- is 55 of 198, 27.8%, and a source a written with an open vowel is 1127 of 1634, 69.0%. The coda is the weak link and is stated rather than buried: the word carries an -m and the character carries the velar -ŋ, a change of place which the corpus permits in 17 of 58 cases, 29.3%, but does not favour. 戶 <i>ga^B</i> → -ga: a source g written with a Chinese g- is 64 of 110, 58.2%, with the vowel open again. The pair is the corpus device for a medial nasal-plus-velar: given a written -ŋ coda followed by a velar onset, the source had a nasal followed by a velar stop in 26 of 29 cases, and the three exceptions are all one word. The whole reading costs 2.26%, about 1 in 44. That is more expensive than the 1 in 37 which retires <i>Bayatur</i> in §8.1 of the paper, and the figure is given here so that a reader can weigh it rather than take the reading on trust. What these characters cannot write. A word with a single velar nasal and no velar stop, which is why <i>tonġa</i> stood here in earlier drafts and has been moved to 車牙若鞮, where two characters write it directly. The <i>Codex Cumanicus</i> forms that appear to fit this frame, <i>tengis</i>, <i>tongus</i>, <i>tangis</i>, are medieval Latin spellings of a single velar nasal, as the Dīwān’s own <i>teñiz</i> and <i>toñuz</i> show for the same words, and they are excluded for the same reason. REASON <i>Tamga</i>, the seal of the ruler, and by extension the officer who keeps it. 當戶 is an administrative office in the Xiongnu hierarchy rather than a ruler’s name, so the register wanted is court vocabulary, and <i>taṃġa</i> is exactly that. The sense is attested, though not under the headword. Kāšġarī’s own entry for <i>taṃġa</i> gives an unrelated sense, a branch of water or an anchorage, but the derivatives in the same dictionary carry the seal throughout: <i>taṃġalamak</i>, “to strike the khan’s seal” (III 353), and <i>taṃġalıġ</i> and <i>taṃġalıċ</i>, “objects bearing the khan’s seal” (I 527). Clauson adds an Uyghur civil document, <i>beg taṃġası elgiñde</i>, “a beg’s seal is in your hand” (TT I 129). So the administrative sense is on record in two independent places, and the later Turkic and Mongol chanceries make <i>taṃġa</i> the ordinary word for the seal, with <i>taṃġacı</i> its keeper. What is not settled. The price, 1 in 44, which is worse than the threshold the paper uses elsewhere, and the reader is told so above. The nasal, which is a change of place rather than a direct correspondence. That the seal sense reaches the headword only through its derivatives, so the chain has one link more than at a row where the dictionary glosses the word itself. And whether the Chinese office 當戶 had anything to do with seals, which the histories do not say and this reading assumes.				

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
闕氏 yanzhi / ezhi	ʔat dʒe ^{B †}	’at ce	Elti	lady; a woman related by marriage
PHONETICS 闕 ʔat → <i>el-</i>: a vowel-initial character whose written -t carries the source liquid. Later Han had no -r or -l in any coda, so a source syllable ending in one was written with a -t or left open: of 37 such syllables in the Sanskrit corpus, 21 were left open and 14 took a -t, 38%. 氏 <i>dʒe^B</i> → <i>-ti</i>: a dental onset with the vowel free. The first character is not polyphonic: Schuessler’s table gives 闕 the single value ʔat, and Baxter and Sagart give Middle Chinese ʔat from Old Chinese *qʰat. The -n that the parallel spelling 焉支 carries belongs to 焉, a different character, which Baxter and Sagart give two readings, both in -n. The second character is another matter: the table gives 氏 as <i>dʒe^B</i>, <i>gie^B</i>, <i>tʂe</i> and <i>kie</i>, and Baxter and Sagart reconstruct *<i>k.deʔ</i> with a velar preinitial, which is where the velar readings come from. This item is character-anchored in the emission row; the second character is less constrained than that alone suggests. REASON Elti, “lady”. Modern Turkish <i>elti</i> is what each of two brothers’ wives calls the other, and in dialect what two wives of one man call each other. It is not only modern: Clauson records <i>élti</i>: glossed “lady”, standing opposite <i>al-jāriya</i> “slave girl”, in a fourteenth-century Arabic-Turkish glossary, which is the sense a consort’s title would want. Nişanyan takes Middle Turkic <i>élti</i> “married woman, as a respectful term” and derives it, with a stated doubt, from Old Turkic <i>élt-</i> “to carry, to lead”. If that holds, the word has the shape of <i>kelin</i> “bride”, which Clauson derives from <i>kel-</i> as “one who comes in (to the family)”: a deverbal noun naming a woman by how she entered it. 闕氏 is the title of the chanyu’s principal consort. Why this replaces the earlier proposal in this supplement. The earlier reading was <i>ece</i> “elder sister”, later <i>eci</i> “senior woman, lady”, both of which are in the Kāşgarî index (l, 86 and l, 87) and the second of which is the better semantic fit. Both discard the written -t of 闕, and the corpus makes that expensive: where the source has a single affricate or sibilant between two vowels, the preceding character was open in 134 of 143 junctions, 93.7%, and carried a -t in 2, 1.4%. That is a worse cost than the 1 in 37 the paper uses in §8.1 against a rival reading elsewhere, so consistency required either conceding it or finding a reading that uses the -t. <i>elti</i> uses it, at 38%. The earlier proposal is recorded here rather than removed, since the reasoning that led away from it is the useful part. What is not settled. <i>elti</i> is not in the Türk Dil Kurumu index nor in the Codex Cumanicus, and the fourteenth-century attestation is in the edition Clauson marks “(only)”. Pulleyblank read the title as Turkic and Mongolian <i>qatun</i>; Vovin rejects that and proposes instead a Yeniseian *<i>alte</i> or *<i>elte</i> “wife”, from Proto-Yeniseian *ʔal(V)it- (Kott <i>alit</i>, Assan <i>alit</i>, Arin <i>biqam-alte</i>), noting that it “could be transcribed in Chinese only as *ʔat-tej or *ʔen-tej”. He is right about the Chinese and the same spelling writes <i>elti</i> by the same steps, so the transcription does not choose between the two languages. §9.1 of the paper sets that out as a measured case of the negative result of §6.1.				
骨都侯 guduhou	kuət ta go	kuit ta go	Kutak	the blessed one
PHONETICS 骨 <i>kuət</i> → <i>kut</i>: default velar onset with the -t coda written, the tightest single syllable in this supplement. 都 <i>ta</i> then begins with the same dental, and Table 8 of the paper measures what that means: at a junction where a written coda is followed by the same-place onset, the source word carries that consonant once rather than twice in 121 of 140 cases (86%). So 骨都 transcribes <i>kuta</i>, not <i>kutta</i>. 侯 <i>go</i> → <i>-k</i> closes it, a Later Han voiced g- standing for a source k more often than for a source g. REASON <i>kut</i> is “kut, uğur, devlet, baht, talih, saadet” in Kāşgarî (heavenly favour, fortune, sovereign grace) and Clauson records the verb <i>kutad-</i> “to be blessed”, the stem behind the <i>Kutadgu Bilig</i>. <i>Kutak</i> itself is not attested: it would be a deverbal or diminutive formation in -(A)k on that stem, which is a live suffix but not one witnessed on this word. Compare <i>Qutalmış</i>, ancestor of the Seljuks of Rum. Note also that <i>kut</i> is the Turkic candidate for 孤塗, where the Chinese gloss 子 “son” fits the Yeniseian proposal better and not this one.				
且渠 juqu	tsa ^{B †} gia	tʂa gia	Çaka	the striker; from çak- ‘to strike fire, to strike a blow’
PHONETICS 且 <i>tʂa^B</i> → <i>ça</i>: ts- for a source affricate is a minority spelling but an attested one, 15% in the Sanskrit corpus and 9% in Ligeti, against a tʂ- character as the default at 37%. 渠 <i>gia</i> → <i>ka</i>: a Later Han voiced g- stands for a foreign k more often than for a foreign g. Both steps are permitted by the measured correspondences. REASON <i>çak-</i> “to strike fire from flint, to sting, to slander”, which Clauson describes as onomatopoeic in character and of violent action, physical or mental. No <i>çaka</i> headword exists in Kāşgarî, the Codex or Clauson, only the verb and its derivatives <i>çakmak</i>, <i>çakır</i>, <i>çakılmak</i>, so <i>Çaka</i> would have to be the -A converb of that stem, and a bare converb serving as an office title is not a demonstrable pattern. The 11th-century emir Çaka Bey of Smyrna (Byzantine Τζαχαῖς, <i>fl.</i> 1081–93) shows that <i>çak-</i> could yield a personal name, but that is a personal name set against a title. The Chinese sources do not record what the office of 且渠 did, so no sense can be tested against a gloss.				
徑路 jinglu	keŋ ^c la ^c	keng la	Kingrak	a large cleaver-like knife

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS 徑 <i>ken^c</i> → <i>king</i>: default velar onset with the -ŋ coda written. 路 <i>la^c</i> → <i>ra</i>: a Chinese l- for a source r-, the direction those counts show in 12 of 14 attestations. The Turkic word's final -k is not written, which Table 7 of the paper licenses at 30%. REASON Not our reading. Friedrich Hirth compared 徑路 with Turkic <i>kıṇrak</i> in <i>The Ancient History of China to the End of the Chóu Dynasty</i> (1908), pp. 65–70, calling it the oldest Turkish word on record. Kāšgarī glosses <i>kıñrak</i> as a knife like a butcher's cleaver, for cutting meat and dough, and the Chinese sources describe 徑路 as a blade used in oath-taking, so gloss and referent agree across unconnected sources. Turkic origin, on Clauson's analysis. Clauson derives <i>kıṇra:k</i> as a deverbal noun from an unattested <i>*kıṇra:-</i>, denominial from <i>kıñır</i>, in the sense of something curved or something that cuts crookedly, so the word is motivated inside Turkic by the standard authority rather than by us. Kāšgarī's index carries the same family (<i>kıñru</i> “askew”, <i>kañrak</i> “palate”, <i>koñur-</i> “to wrench, bend back”), and the Codex glosses <i>Kingir</i> as <i>curvus</i>. Clauson also records the word alive in Teleut as <i>kıñırak</i> and Kirghiz <i>kıṇarak</i> / <i>kıñırak</i>, a rough two-edged knife for cutting felt and scraping hides. Two cautions. Kāšgarī's text at III 382 reads <i>kırpa:k</i>, corrected in the margin to <i>kıṇra:k</i>, so the form rests on an emendation; and of the family it is <i>kıñır</i> that fits the transcription exactly, while <i>kıñrak</i>, the member meaning a knife, needs the unwritten final -k allowed at 30%. Iranian origin. The object has also been derived from an Iranian word for a short sword, the <i>akinakes</i> type, historically plausible, since the Xiongnu's western neighbors were Iranian-speaking. But 路 is a liquid-onset character, so the channel requires a liquid in second position, and the akinakes frame (k–n–k) has none. As stated, that account fails on the frame. Two cautions there too: we have not seen the exact Iranian etymon its proponents advance, and the channel shows what a transcription permits, never what it requires. §9.1 of the paper sets both origins out.				
服匿 funi	buk [†] ɳɿk	buk nɿk	Buknik	clay jar (kūp)
PHONETICS 服 <i>buk</i> → <i>buk</i>: exact, including the written -k coda. 匿 <i>ɳik</i> → <i>nik</i>: retroflex ɳ for n is routine and the second -k is also written. This is the closest fit to its own transcription of anything in this supplement: no segment inserted, none discarded. REASON A clay jar (Turkish <i>kūp</i>). One of the few Xiongnu material terms the Chinese sources preserve, which is what makes it worth a reading at all.				
甌脫 outuo	ʔo t ^h uat	’o tuat	Otuyat	the border watchtowers
PHONETICS 甌 <i>ʔo</i> → <i>o</i>: vowel-initial, direct. 脫 <i>t^huat</i> → <i>tuyat</i>: the aspirated t- is direct and the -t coda is written; the only addition is the medial -y-. REASON The border watchtowers, or the empty zone between Xiongnu and Han territory, for which the Chinese sources use 甌脫.				
胡祿 hulu	gɔ [†] lok	go lok	Külüg	famous, renowned
PHONETICS 胡 <i>gɔ</i> → <i>kü</i>: the devoicing is normal, a Later Han voiced g- standing for a source k more often than for a source g; ü cannot be written and surfaces as o. 祿 <i>lok</i> → <i>lüg</i>: the -k coda is written. The consonant frame k–l–k is secure and only the vowels are free. REASON <i>külüg</i> “famous, renowned” is an Old Turkic epithet of praise, recorded by Clauson; note that the form is <i>külüg</i>, not <i>külük</i>, and it is not a headword in either lexicon used here. Two objections stand. First, the Chinese-side record gives the meaning of 胡祿 as “quiver”, not a term of praise, though that gloss is itself marked low-confidence and chronologically shaky. Second, the reading the gloss wants is barred by the transcription: the one Turkic quiver word whose skeleton fits is <i>okluk</i> (Kāšgarī, glossed <i>sadak</i>), and it is vowel-initial, whereas 胡 carries a voiced velar. In the Sanskrit corpus a source vowel-initial syllable was written with a zero or glottal onset in 96% of 183 cases, and with a velar in none. So this is an item where the Chinese gloss names a target, a Turkic word exists for it, and the spelling rules that word out, recorded because a negative of that shape is worth more than a resemblance.				

TABLE S5 The titles and the handful of common nouns the Chinese histories preserve, in the order of Appendix A. The lexical items are the more interesting group because some of them are glossed in the source, which means a proposal about them can in principle be checked.

S1.10 Ethnonyms

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
匈奴 Xiongnu	huoŋ na	huong na	Hung	the ethnonym, from the first character alone
PHONETICS 匈 <i>huoŋ</i> → <i>hung</i>: direct, with the -ŋ coda written. 奴 <i>na</i> is discarded rather than read, on the assumption that 奴 “slave” is a Chinese pejorative appended to the name rather than part of it. REASON Dropping 奴 is common in the literature and what is proposed is only the vocalisation of the remaining syllable.				
羯 Jie	kiat	kiat		no reading proposed; kıyat withdrawn

FORM, AS WRITTEN IN THE HISTORIES	LATER HAN READING OF THOSE CHARACTERS (SCHUESSLER)	THE SAME READING IN TURKISH ORTHOGRAPHY	READING WE PROPOSE	SENSE WE PROPOSE FOR THAT READING
PHONETICS A single character, <i>kīat</i>, with the <i>k</i>- default and the <i>-t</i> coda written, so the consonant frame <i>k...t</i> is secure. The vowel carries information, and step 53 counts it. Characters whose Schuessler vowel contains <i>ia</i> write a source a in 28 of 28 cases; a source e is 2.0% of the corpus and appears in none of them. On the Later Han layer this character writes a source a. That measurement is recorded here and deliberately not used as an argument. The Jie appear in the fourth century CE and the corpus behind the figure is Later Han, which is the period mixing §7 of the paper warns against; and Proto-Yeniseian <i>*keʔt</i> is a reconstruction of a proto-form, so it does not fix the vowel of the fourth-century word and the measurement is not aimed at what Vovin actually claims. The one-syllable words the frame admits are <i>kat</i> “layer” and <i>kad</i>, which Kāšgarī glosses “kar fırtınası, insan öldüren bora, tipi”, a blizzard or a killing gale. Neither is an obvious ethnonym. REASON No reading is proposed here. An earlier draft read this as <i>kīyat</i>, “of flooding”. That is withdrawn, and the reason is not a fine judgement: <i>kīyat</i> is not a headword in the Kāšgarī index, the Codex or the <i>Irk Bitig</i> glossary, and the root it would be built on means the opposite of what was claimed. Kāšgarī’s <i>kīy-</i> family is <i>kıymak</i> “sözden dönmek; kıymak, eğrilemesine doğramak”, to break one’s word and to cut crosswise, with <i>kıyık</i> “cayma, caymak” and <i>kıyturmak</i> “iğrilemesine kestirmek”. Nothing in it is about water. The sense offered had no source. The reading in print stands unopposed. Pulleyblank and Vovin read the ethnonym as <i>*ket</i> and connect it with Proto-Yeniseian <i>*keʔt</i> “person, human being”, the Yeniseian self-designation, reflected in Ket <i>keʔt</i>, Yug <i>keʔt</i>, Kott <i>hit</i>, Arin <i>kit/qit</i> and Pumpokol <i>kit</i> (Vovin 2000, <i>CAJ</i> 44/1, p. 91). This is the only item in the record where a reading from the literature stands with nothing of ours beside it, and it is left that way. Why the vowel count is not used against it. It would tell against <i>*ket</i>, and this author has an interest in its telling against <i>*ket</i>. That is the reason to state the two objections above rather than the conclusion, and to leave the row to the reading that has a proposer and a family of cognates behind it.				

TABLE S6 Group names rather than personal names, and subject to a further difficulty: a Chinese exonym need not transcribe anything the group called itself.

s1.11 Characters with more than one Later Han reading

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
鞬	te	de	且鞬侯 壺衍鞬 握衍胸鞬 若鞬 復株累若鞬 搜詣若鞬 車牙若鞬 烏珠留若鞬 烏累若鞬 呼都而尸道皋若鞬 攣鞬
犁	lei	li	响犁湖 撐犁孤塗單于 撐犁
且	tsa ^B	ts ^h ia ^B · ts ^h ia · tsia	且鞬侯 且渠
呼	ha	ha ^c	呼韓邪 呼都而尸道皋若鞬
單	džan	tan	單于 撐犁孤塗單于
累	lui	lyai ^B · lyai ^c · luai ^B	復株累若鞬 烏累若鞬
衍	jan ^B	jan ^c	壺衍鞬 握衍胸鞬
冒	mək	mou ^c	冒頓
响	huo	huo ^c	响犁湖
復	buk	bu ^c	復株累若鞬
搜	so	šu	搜詣若鞬
支	tše	kie	鄯支
曼	man ^c	man · muan ^c	頭曼
服	buk	bu ^B	服匿
氏	dže ^B	gie ^B · tše · kie	闐氏
當	taŋ	taŋ ^c	當戶
稽	k ^h ei ^B	kei	稽粥
耆	gɨ	tśi ^B · ki ^B	屠耆
而	ńə	ńɨ	呼都而尸道皋若鞬
胡	gɔ	ga	胡祿
臣	džin	gin	軍臣

CHARACTER	READING PRINTED	OTHER READINGS IN THE TABLE	ITEMS IN WHICH IT APPEARS
蠡	le ^B	loi ^B	谷蠡
谷	kok	jok · lok	谷蠡
車	tʃ ^h a	kia	車牙若靺
連	lian ^B	lian	虛連題
道	dou ^B	dou ^C	呼都而尸道皋若靺
邪	ja	zia · zia	呼韓邪
頓	tuən ^C	tuət	冒頓
題	de	de ^C	虛連題

TABLE S7 Every character in the record for which Schuessler's table gives more than one Later Han reading, with the reading printed in Appendix A of the paper and the alternatives not used. Generated from [data/derived/polyphony.csv](#), which the numbered script step 24 writes, so this table cannot drift from the repository. The daggers in the tables above mark these same polyphonic characters. Where an alternative would change a proposed reading, the entry for that item says so.

S1.12 The two reconstructions of each character compared

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
粥	tʃuk	t-quk	palatal → velar	—
軍	kun	[k] ^w əər	—	dental nasal → liquid
臣	dʒin	[g]i[ŋ]	palatal → velar	dental nasal → velar nasal
稚	ɖi	lɾəəj-s	dental → liquid	—
維	wi	ɡ ^w ij	glide → velar	—
詹	tʃam	[t]am	palatal → dental	—
衍	jan	N-q(r)anʔ	palatal → velar	—
權	gyan	[g] ^w rar	—	dental nasal → liquid
韓	gan	[g] ^ʳ ar	—	dental nasal → liquid
邪	ja	səə.ɡA	palatal → velar	—
支	tʃe	ke	palatal → velar	—
若	ɳak	nak	palatal → dental	—
車	tʃ ^h a	[t.q ^h](r)A	palatal → velar	—
而	ɳə	[g]ew	palatal → velar	—
尸	ʃi	ʃ[əə]j	palatal → liquid	—
道	dou	l ^ʳ uʔ-s	dental → liquid	—
單	dʒan	dar	palatal → dental	dental nasal → liquid
于	wa	ɡ ^w (r)a	glide → velar	—
塗	da	l ^ʳ a	dental → liquid	—
闕	ʔat	q ^ʳ at	glottal → velar	—
氏	dʒe	k.deʔ	palatal → dental	—

CHARACTER	LATER HAN (SCHUESSLER)	OLD CHINESE (BAXTER & SAGART)	ONSET CLASS	CODA CLASS
路	lɑ	Cæ̃.rʰak-s	—	open → velar stop

TABLE S8 The 22 characters of the record, of the 60 present in both reconstructions, where the two give a different class of segment in at least one position; the other 38 agree. This is the measurement §11 of the paper reports, at 37%. It is not a correction to Appendix A: Baxter and Sagart reconstruct a stage several centuries before the transcriptions and Schuessler one several centuries after, so the two bracket the moment of writing rather than dating it, and this table is the width of that bracket. Their *-s and *-ʔ are treated as having become tones by the Han and are not counted as codas, and a trailing *-j or *-w is read as the offglide of a diphthong rather than as a consonant. Generated from [data/derived/period_depth.csv](#), which the numbered script step 43 writes.

S1.13 How to disagree

The repository includes [src/lookup_lexicons.py](#), which searches Kāšgarī's *Dīwān*, the *Codex Cumanicus* and any EPUB lexicon placed beside them, folding the spelling conventions that separate Turcological from Turkish orthography (q ≡ k, ñ ≡ ng, ş ≡ s, ı ≡ i). Type any proposed form into it and see what the medieval sources actually have. That is how the *Codex* attestation for *inak* was found, and how the -tI count that ruled out the competing suffix was made.

For the reader who disagrees with our proposed names, we suggest disagreeing not with "this is unlikely", since most of them are, but with "this word is not attested before date X", or "this suffix does not attach to that stem", or "the Chinese character cannot carry that coda". Those are the objections that would be useful for the author.

S1.14 Status

Thirty-seven of the forty readings in Tables S2 to S6 carry the same label in the machine-readable file: *author's proposal; the paper measures the steps it takes for the name but does not present it as a finding*. That label is the honest summary of them, and it is not weakened by the fact that some are more attractive than others. Three rows carry a different status, and §S1.1 names them: 若鞮 as *Īnakt*, which the paper argues and which meets its standard; 徑路 as *Kingrak*, a comparison taken from the literature rather than made here, which the paper then tests; and one compound that is read through its constituent rows.

Twelve rows changed status while this supplement was being written, and the changes are recorded here rather than quietly absorbed. 孤塗 and 谷蠡 had no proposal at all in earlier drafts and now carry one. 冒頓 was argued in the paper in an earlier draft and is now an author's proposal like the others: §9.2 of the paper prices the three values its second character can take and does not choose among them, and the proposal made here is set out in S1.4 with its objections. 屠耆 was carried as a comparison taken from the literature and tested in the paper, and is now an author's proposal like the others: the reading could not be traced to any published source, and the paper no longer argues it. 闕氏, 單于 and 軍臣 carry different readings from the ones earlier drafts gave them, in each case because a measurement made since told against the earlier reading; the entries state what the measurement was. At 軍臣 it was that a source voiceless affricate written with a voiced dʒ- is 0 of 53, which retired *Künçin* and sent the reading to the second of the character's two tabulated values. At 狐鹿姑 the reading itself is unchanged, but two objections the entry raised against it are withdrawn: leaving a liquid unwritten before an open character is the majority spelling and not an impossibility, and a written coda reaching forward to the next onset is the commonest use of such a coda. Both had been judged by ear before they were counted. At 羯 the proposal was withdrawn outright and none put in its place: *kıyat* is not a headword anywhere and the root it needed means to cut and to break one's word, so the sense offered had no source. That row now leaves the field to the reading in print, and §S1 records why a measurement that would have told against that reading is reported there but not used. At 呼都而尸道皋若鞮 the proposal was withdrawn too, and for measurements rather than for want of a source: a source vowel-initial syllable takes a Chinese glottal onset 137 of 140 and a velar 0 of 140, and both elements of the reading sat on velar characters; no character in the name carries a liquid, where a source l- takes a Chinese l- 262 of 288; and the reading left four of the eight characters carrying nothing, where no pair in the 1,110-pair Sanskrit corpus uses more than one character beyond its source syllables. *Uluğ* keeps its place at 烏累若鞮, which has a glottal character for the vowel-initial syllable and a liquid character for the -l-. That is the point of keeping the file: the same word was proposed in two rows, and the transcription, not the sense, said which one could hold it. At 車牙若鞮 it was that a source vowel-initial syllable written with a Chinese ŋ- is 0 of 140, which retired *Çöge-Ok*; the reading that replaces it, *Tunga*, was carried at 當戶 in earlier drafts and is moved here because those characters write a nasal followed by a velar stop and this word has a single nasal. 當戶 carries *Tamga* in its place, with its price stated in the entry.

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